



Smart Fire Detection and Response System

Automated fire detection, targeted water suppression, and safety monitoring using IoT



أعربت وكالة ناسا عن قلقها
المتزايد ازاء حركة اجسام
غريبة حول الارض ليتبين لاحقاً
هم اقمار جديدة للأرض واحد
في السماء و 3 هم لجنة
التقييم
لمعرفة اسمائهم في
السلاليد القادم

أقمار الأرض الجديدة



د. سامح العطري



شكراً لكم على جهدكم معنا
خلال هذا الفصل



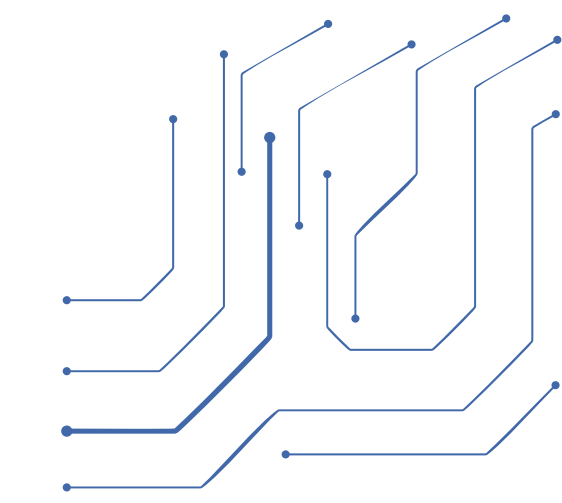
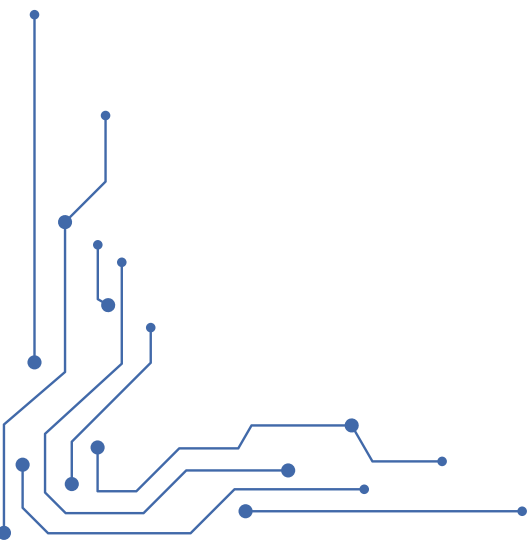
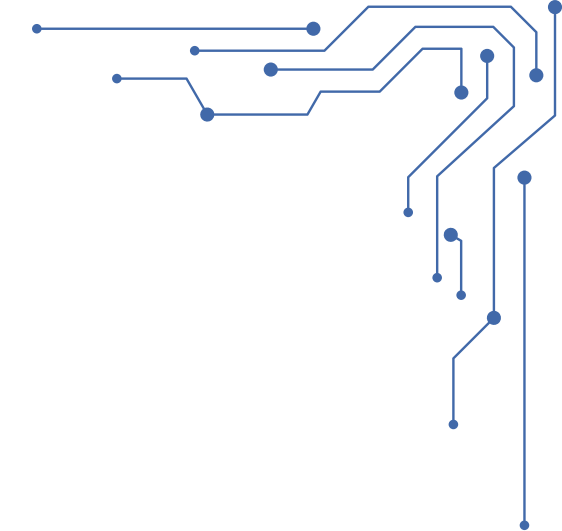
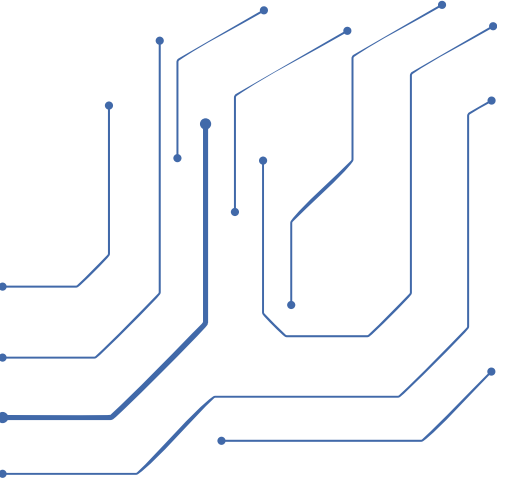
أ. الحسين

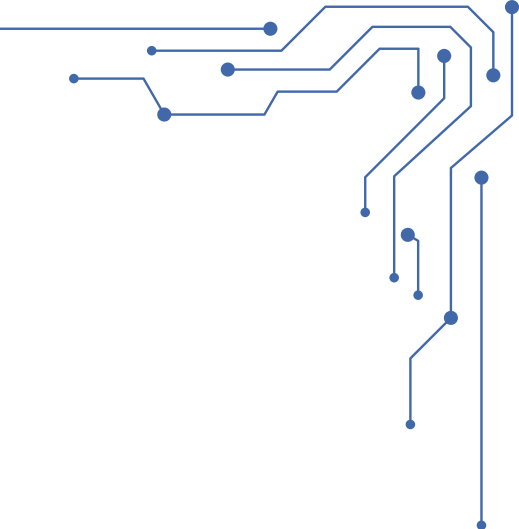
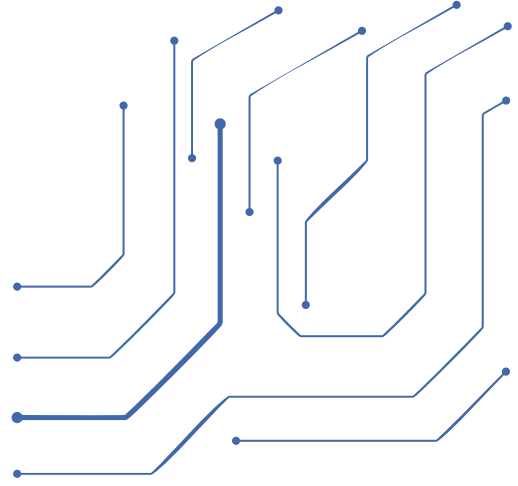


أ. أحمد

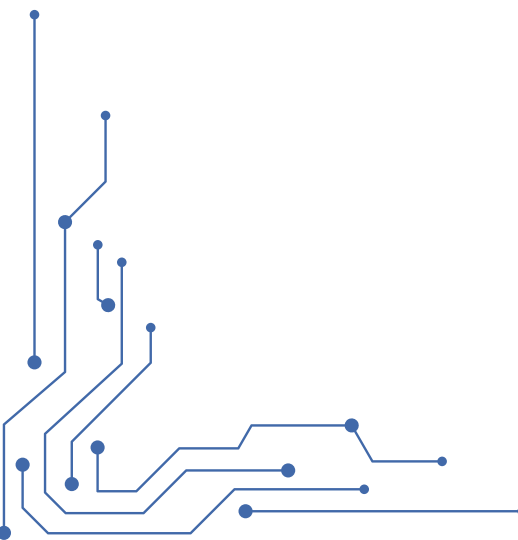
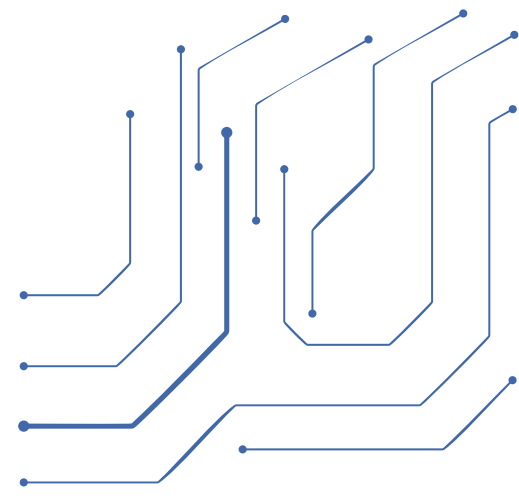
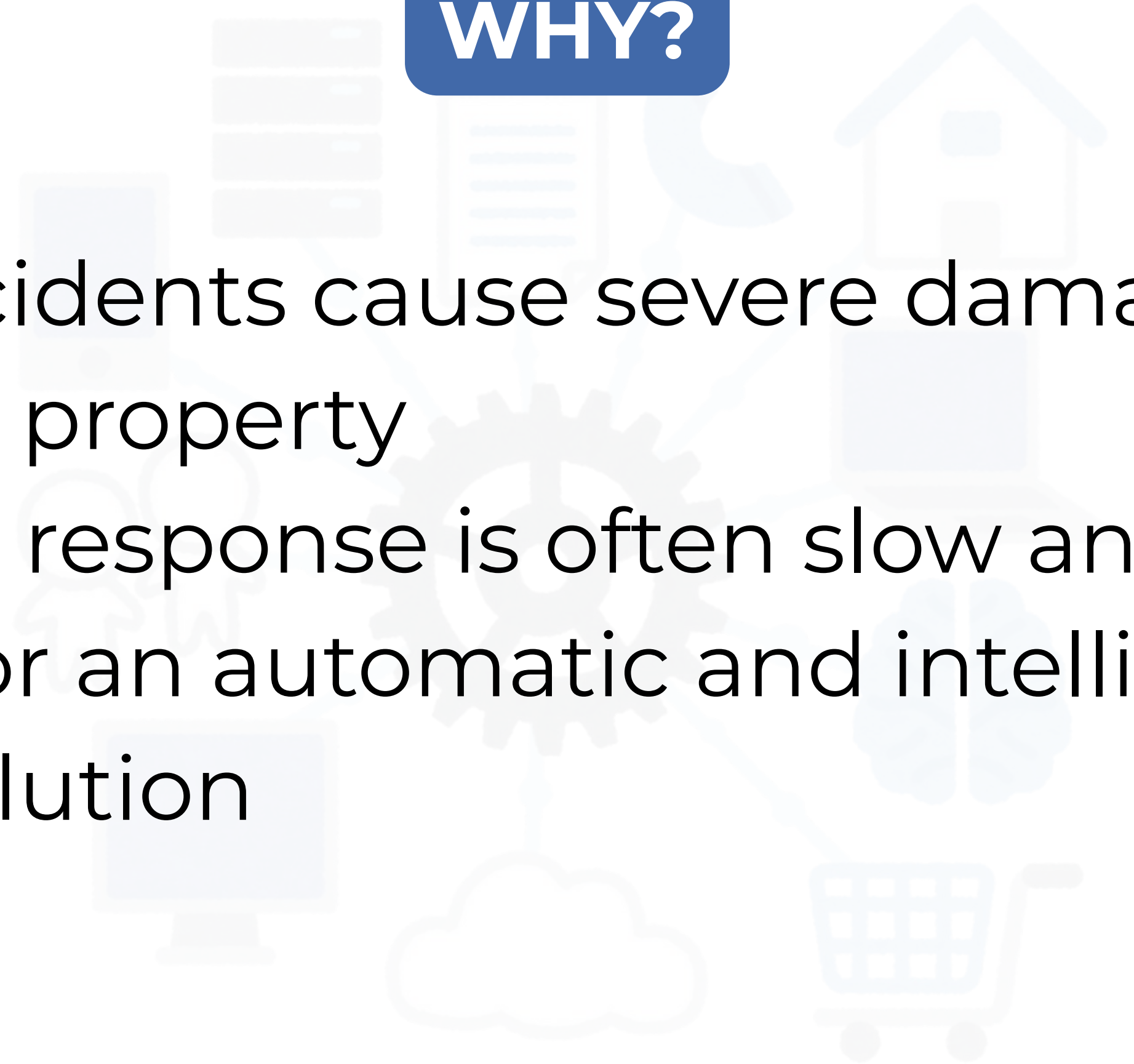
Project Team

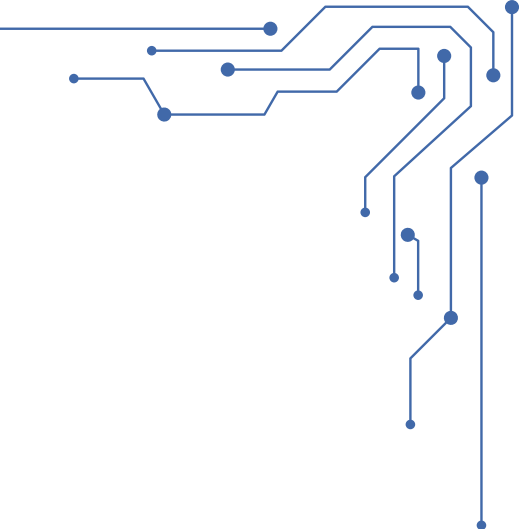
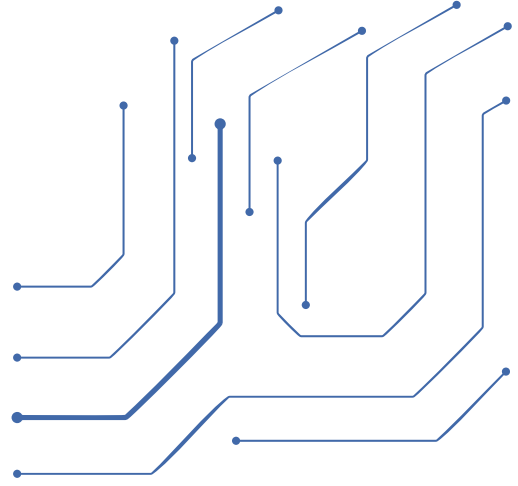




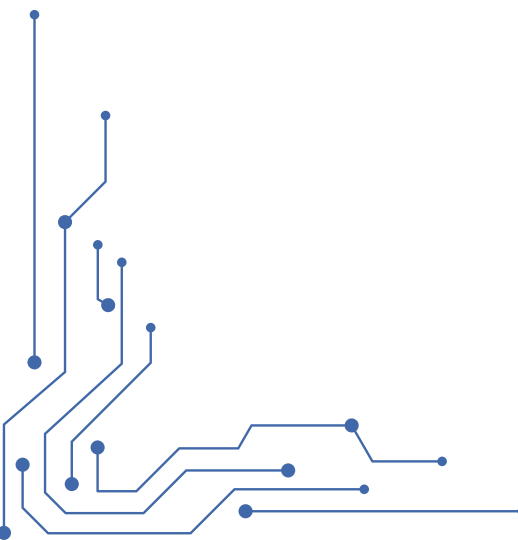
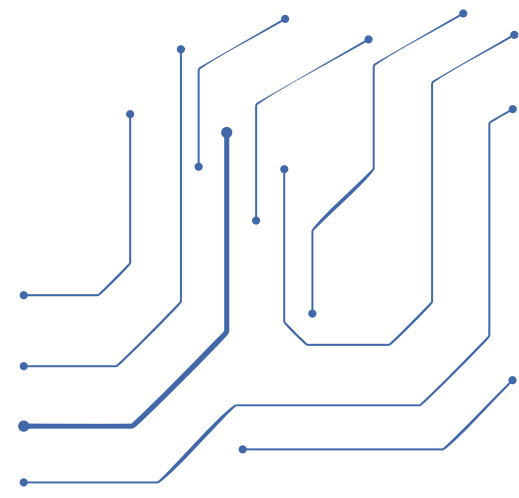


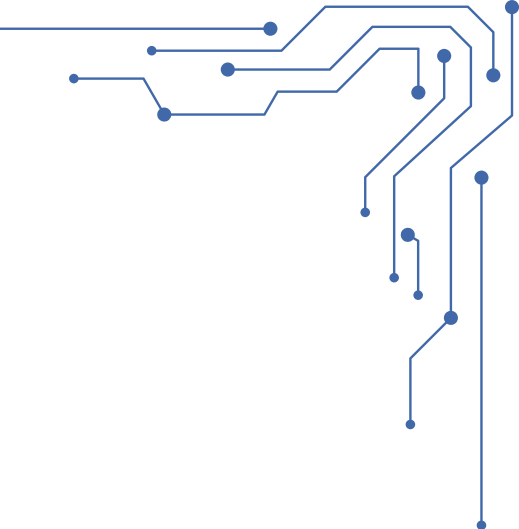
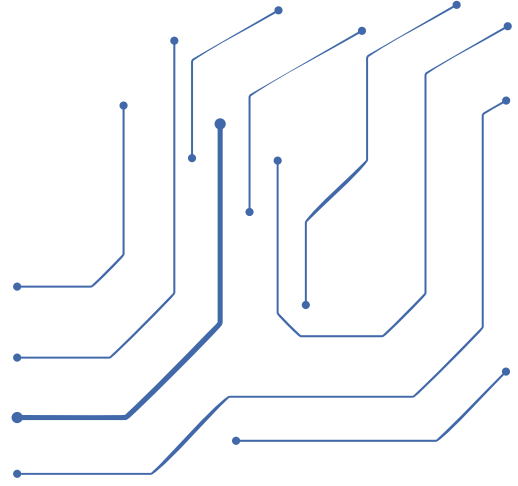
WHY?

- Fire accidents cause severe damage to lives and property
 - Manual response is often slow and unsafe
 - Need for an automatic and intelligent safety solution
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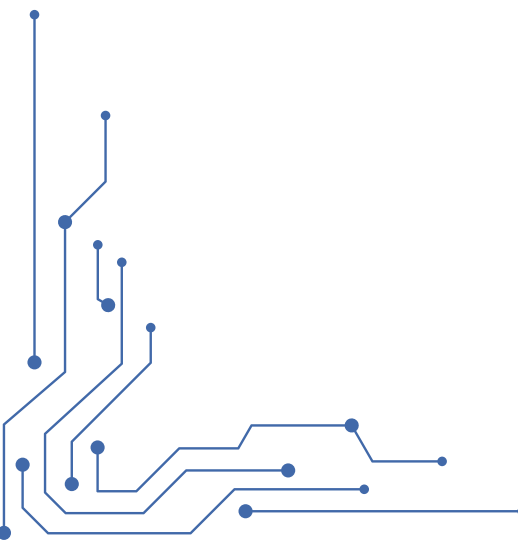
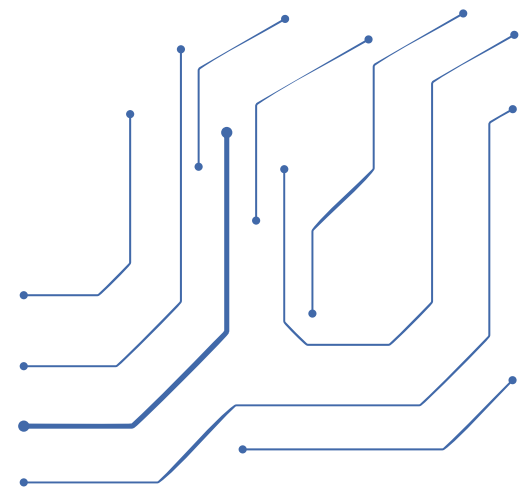


WHAT

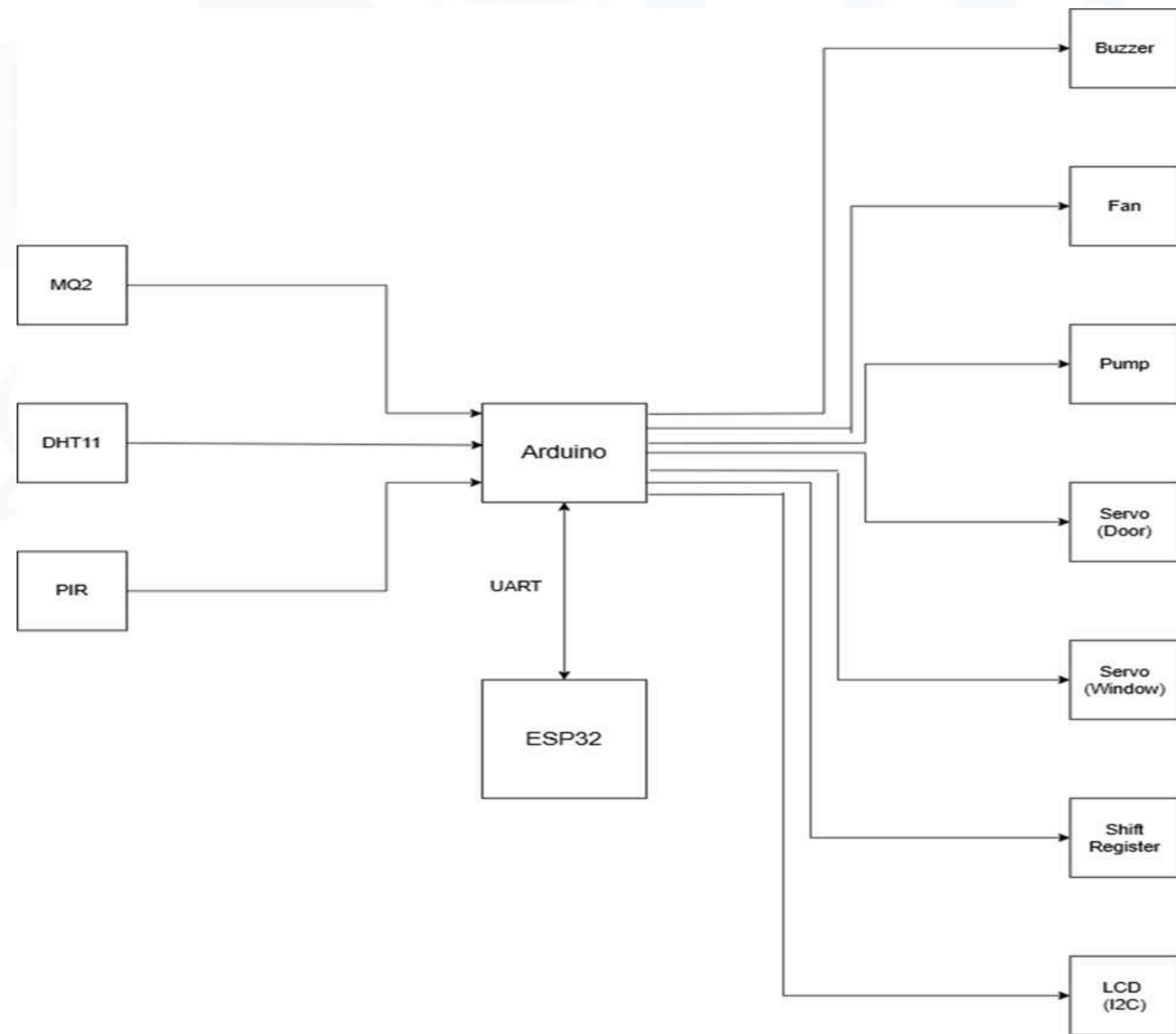
- Smart fire detection system for shops and factories
 - Detects fire and smoke
 - Responds automatically
 - Monitors human presence
 - Provides remote notification
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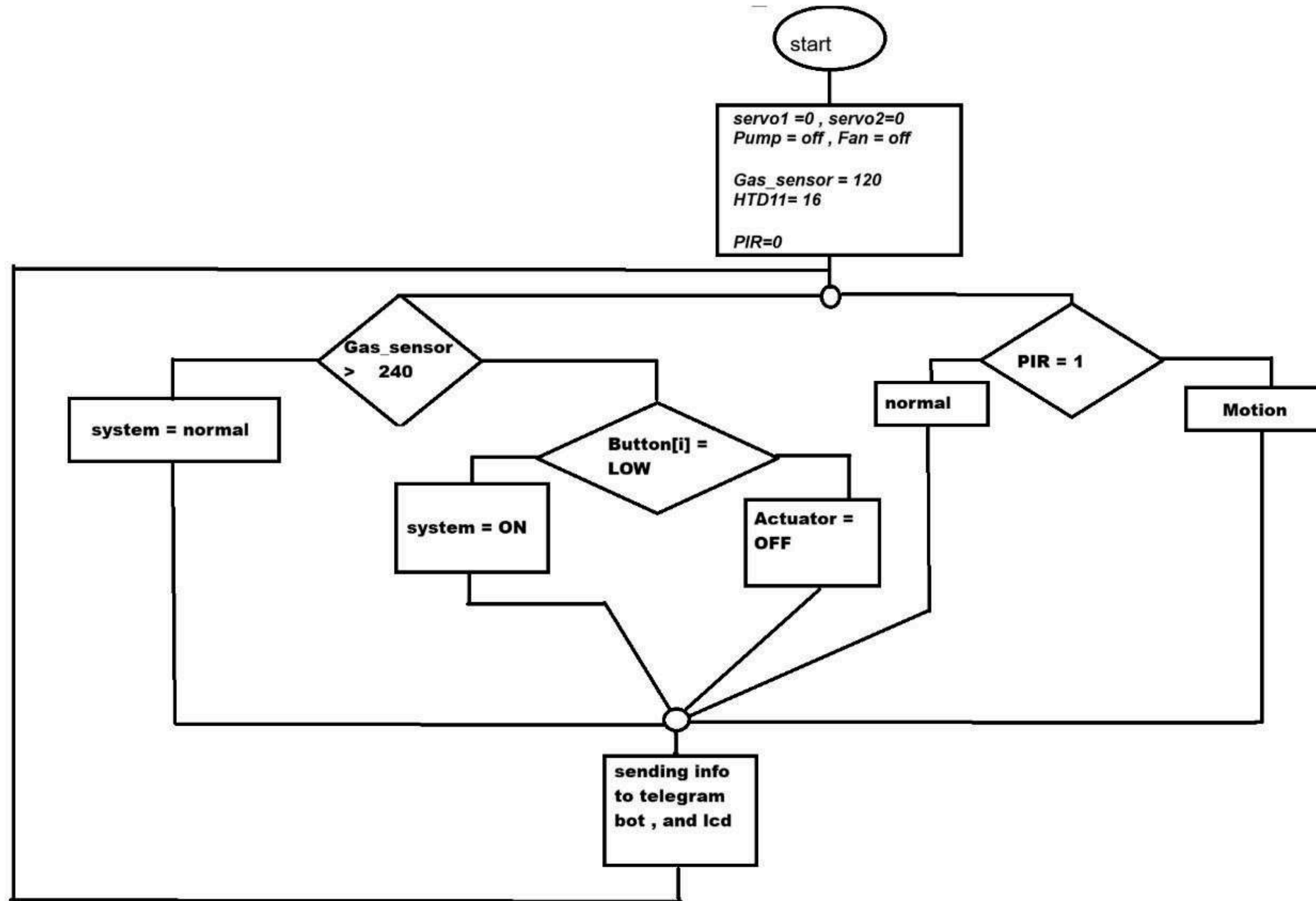
HOW

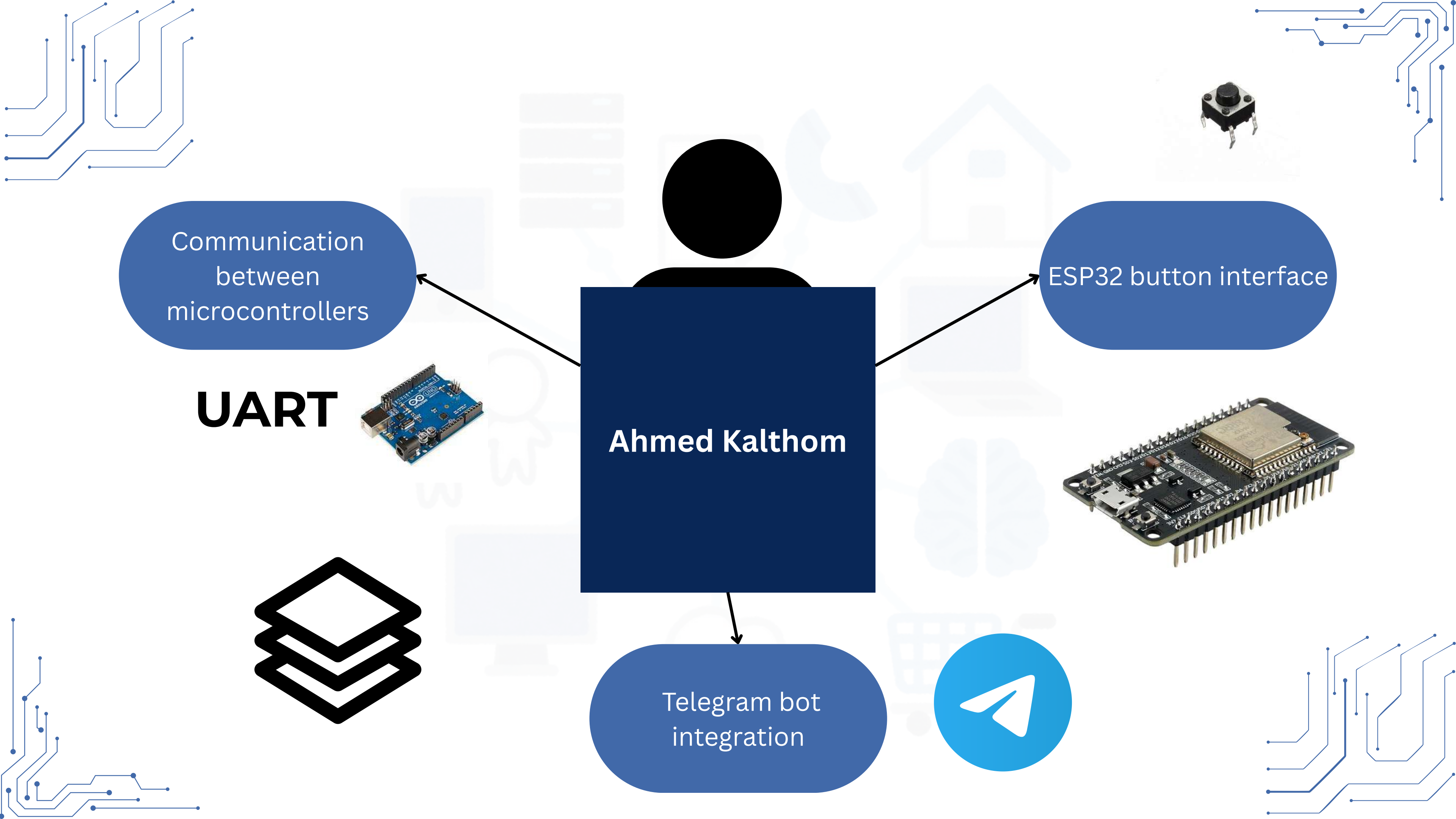
- Sensors detect fire, smoke, and human presence
 - Arduino and ESP32 process data
 - Water pump targets fire direction
 - Fan removes smoke
 - Servo motors simulate doors and windows
 - Telegram bot sends alerts
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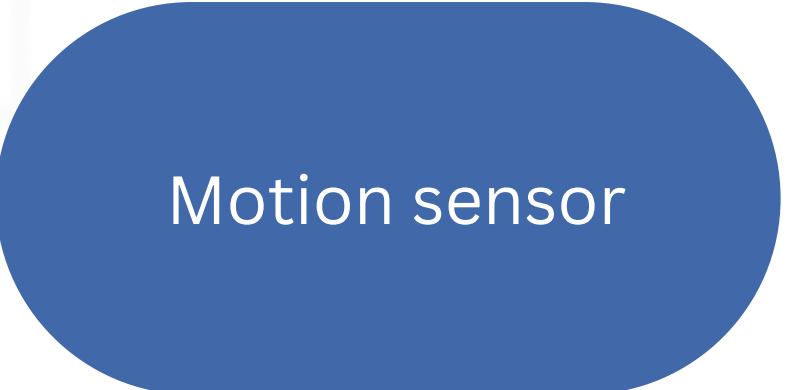
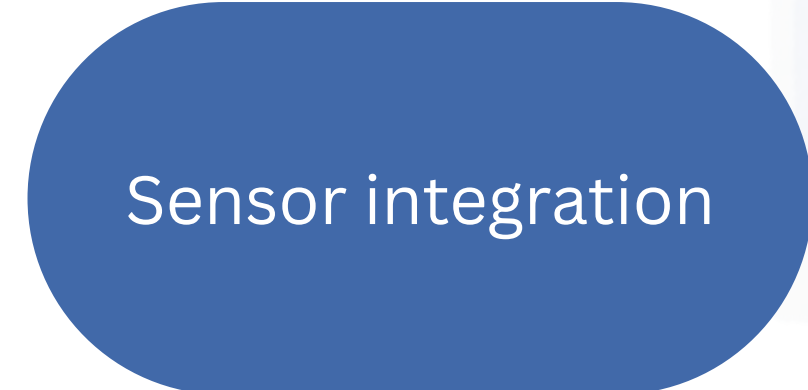
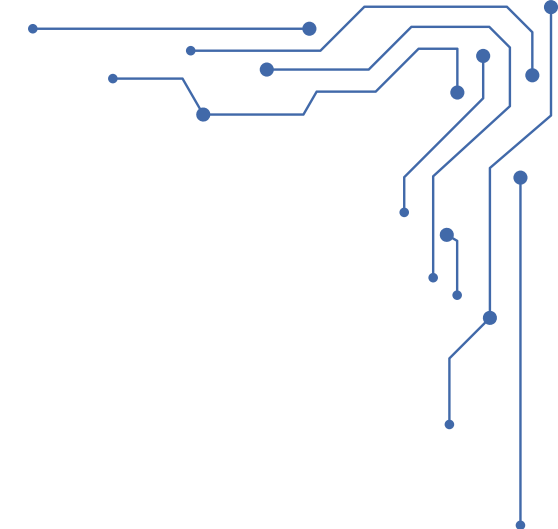
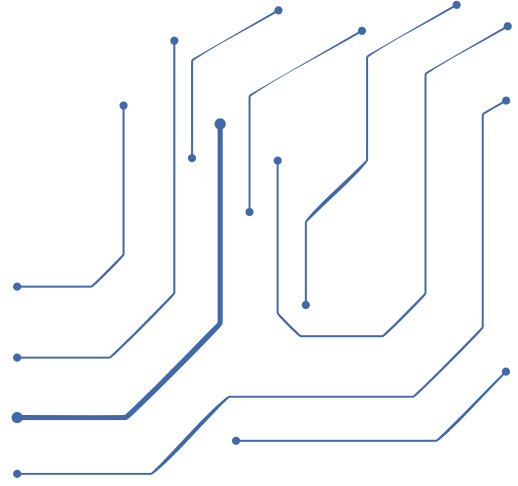
Block Diagram



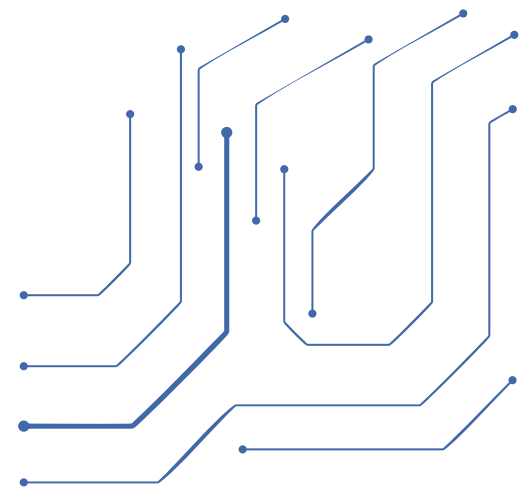
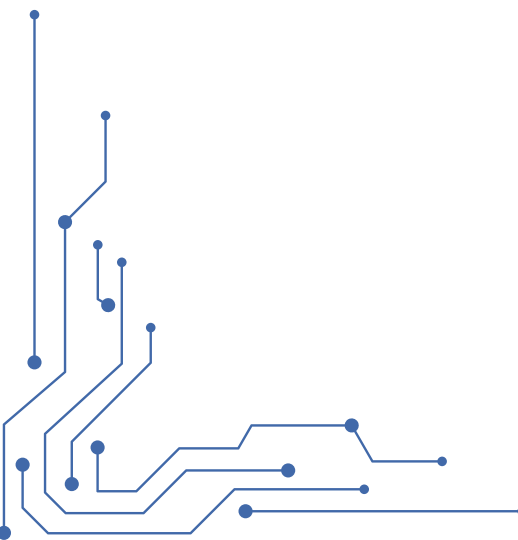
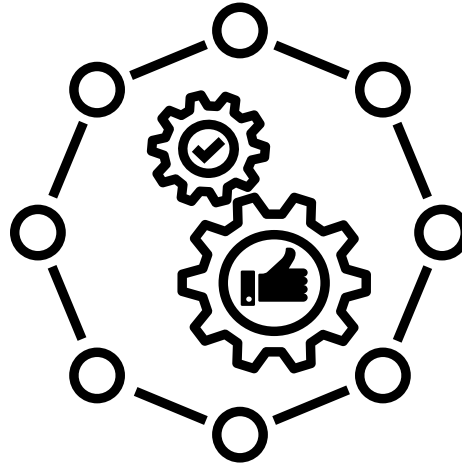
Flow chart







ahmed alrabee

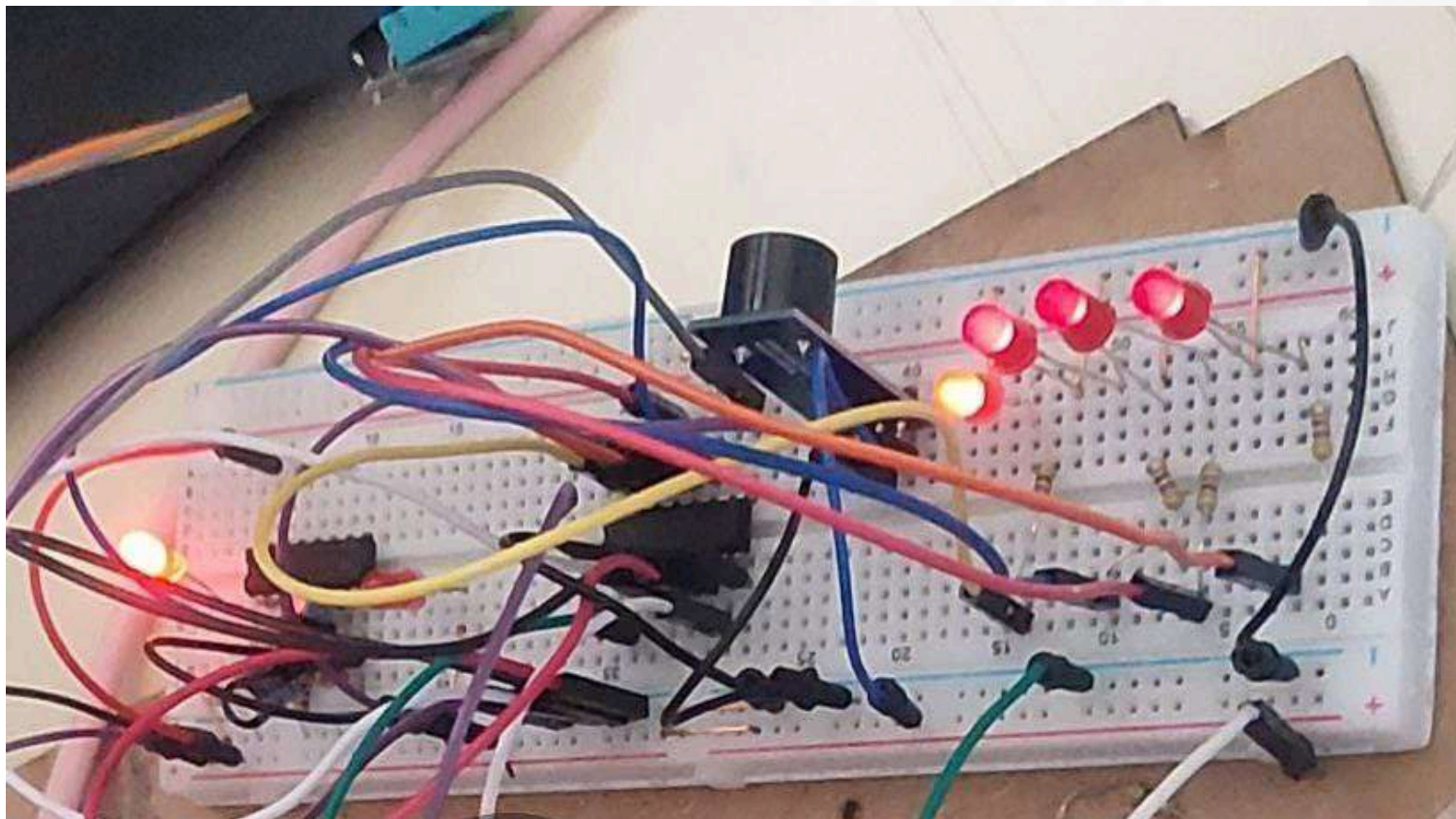


Ammar Al Jodeh

Buzzer integration

Shift register for visual
alarm

Clock pulse circuit



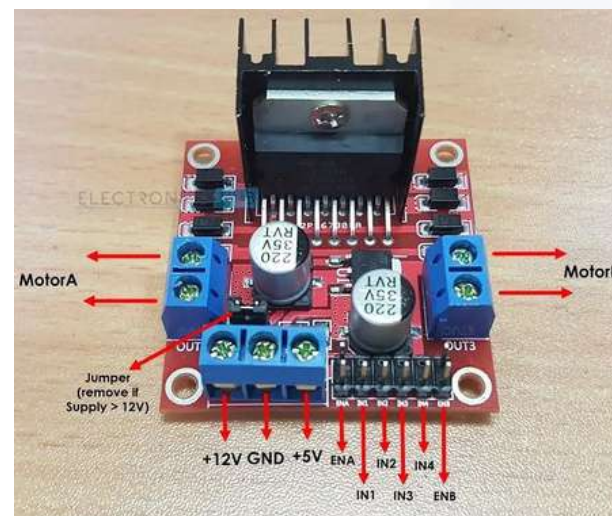
LCD display

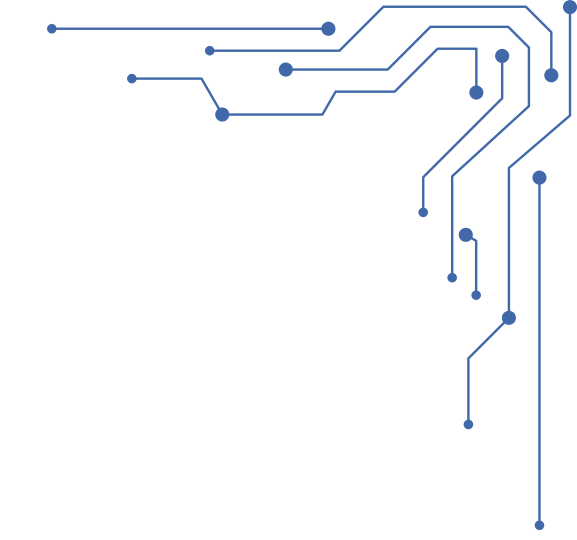
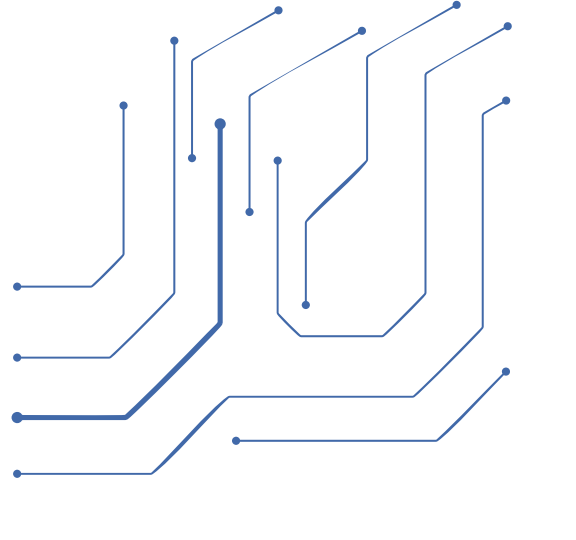
Relay control

Anwar Alshahme

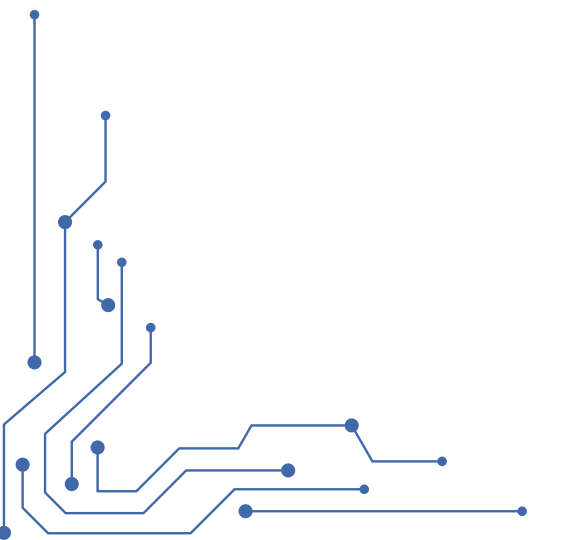
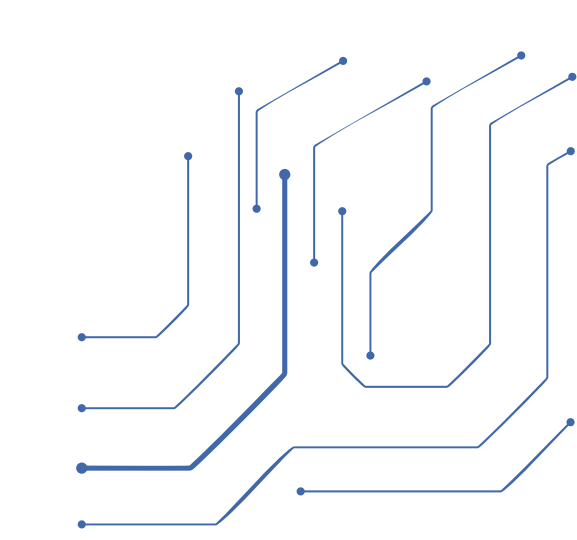
door / windows control

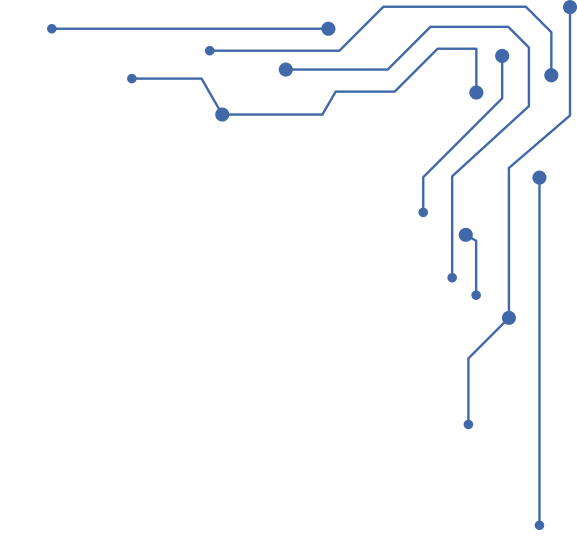
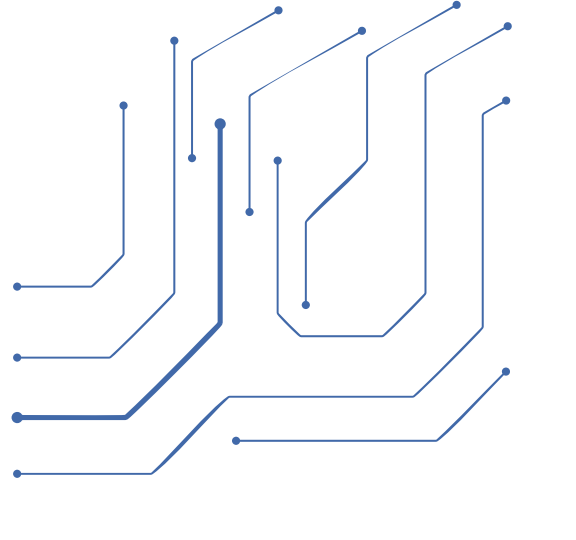
I2C



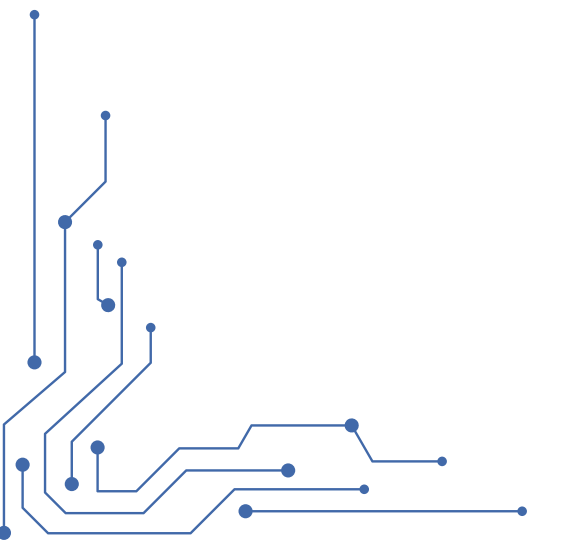
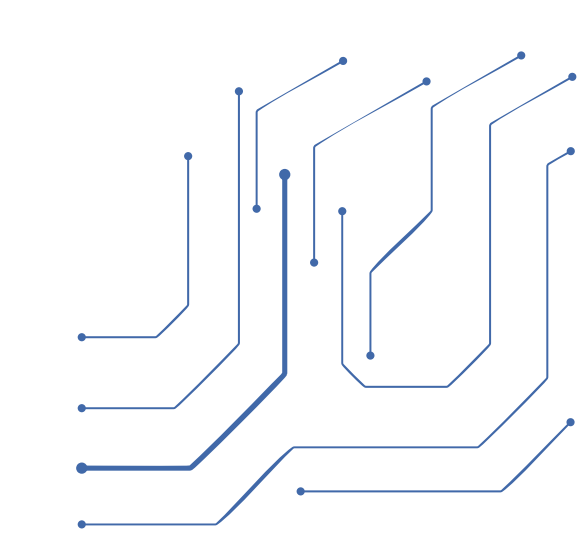


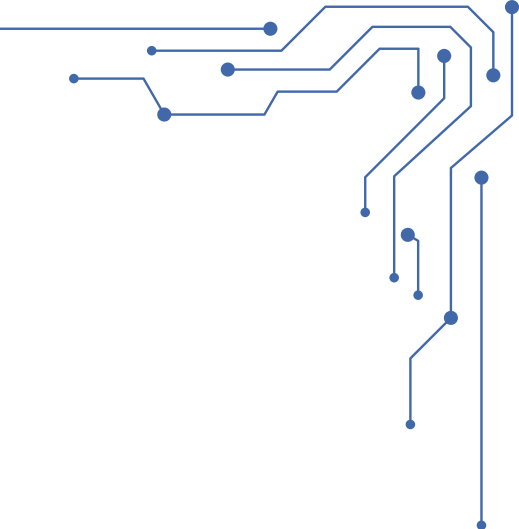
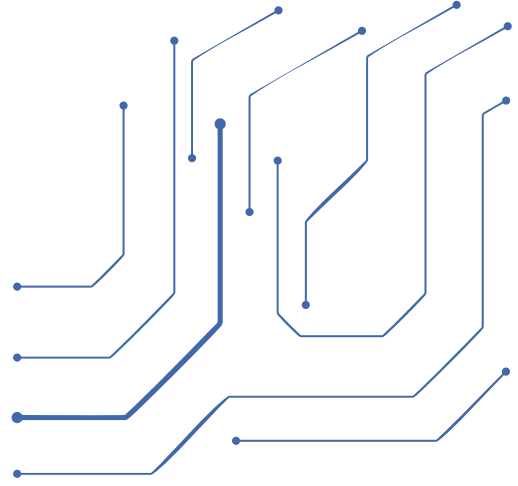
Challenges and Issues

- Magnetic interference caused by relays and driver circuits
 - UART communication instability between controllers
 - Power supply and voltage regulation issues
 - Hardware integration and compatibility problems
 - Low quality of locally available electronic components
 - Team coordination and workload imbalance
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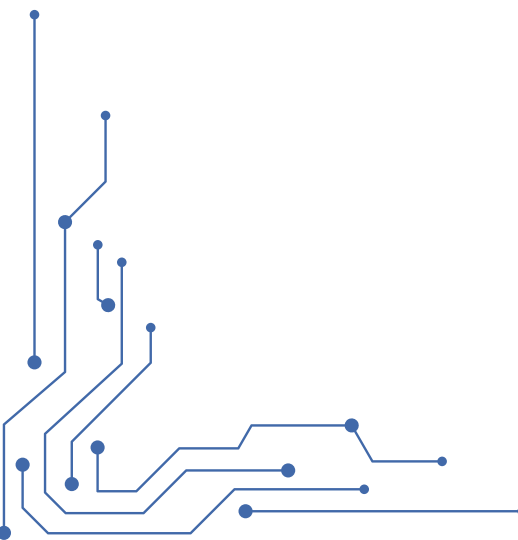
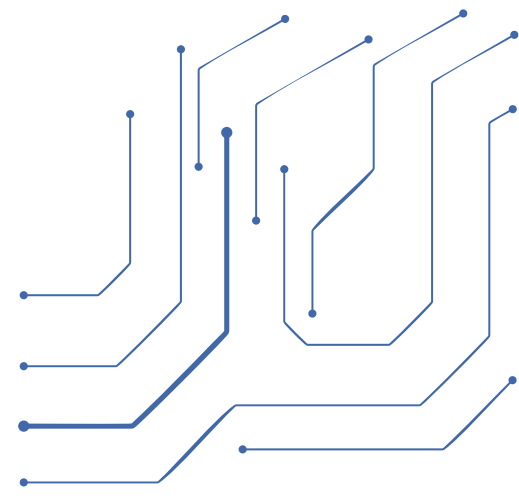


Advanatges

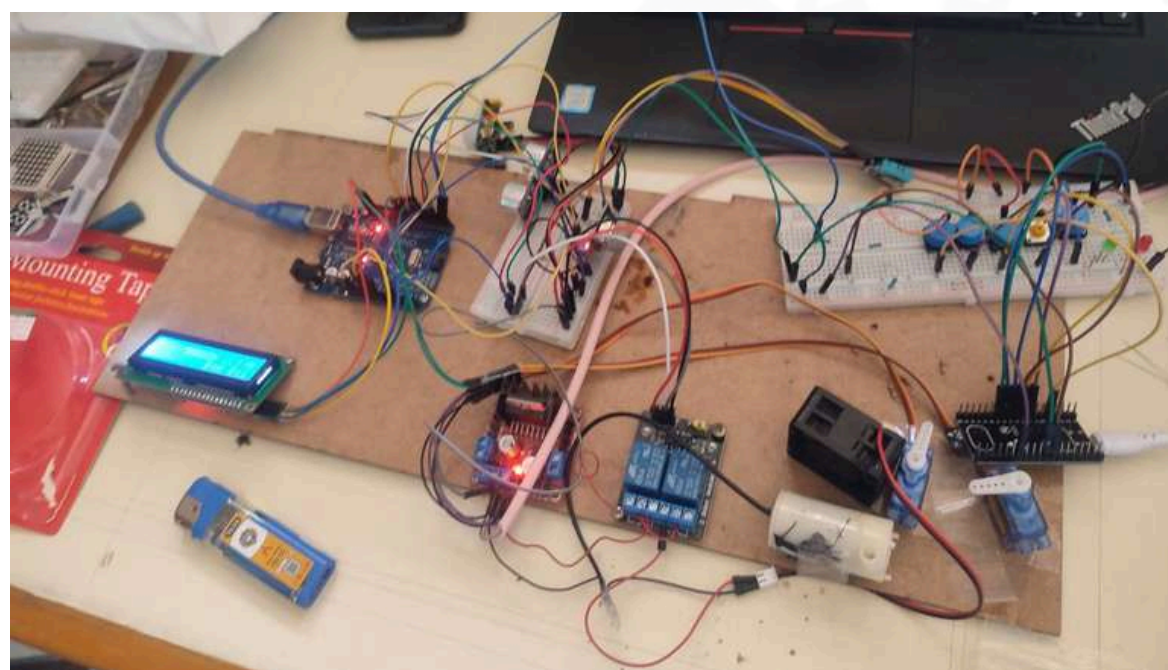
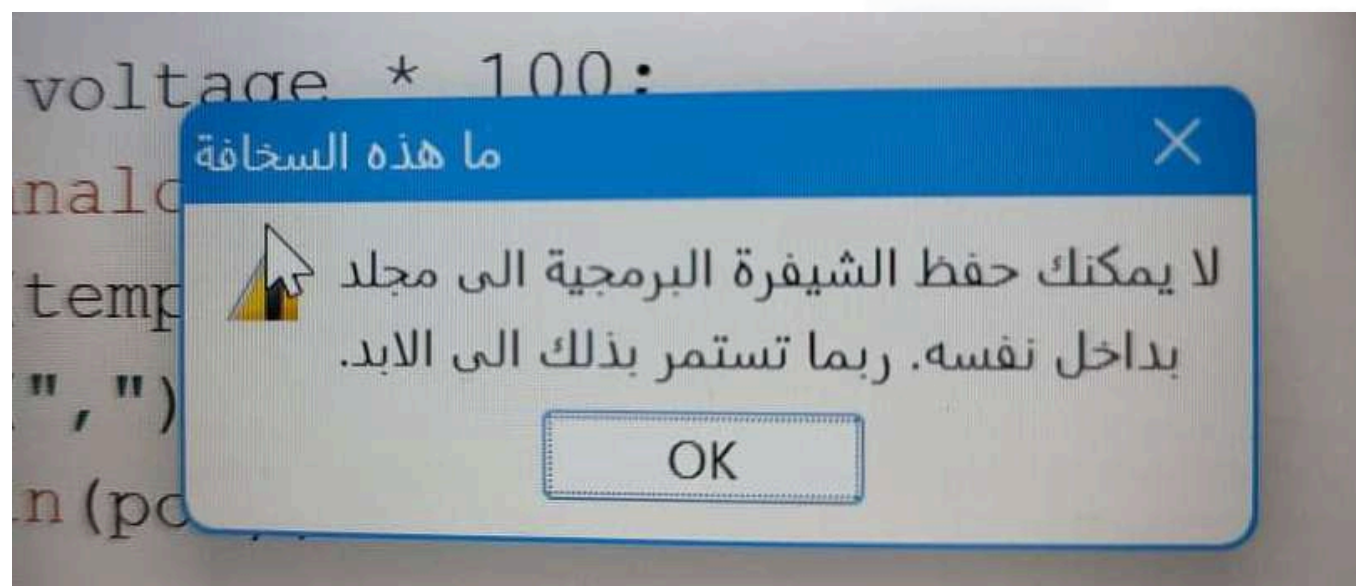
- Automatic fire response
 - Directional water spraying
 - Human presence detection
 - Remote monitoring via Telegram
 - Manual control option
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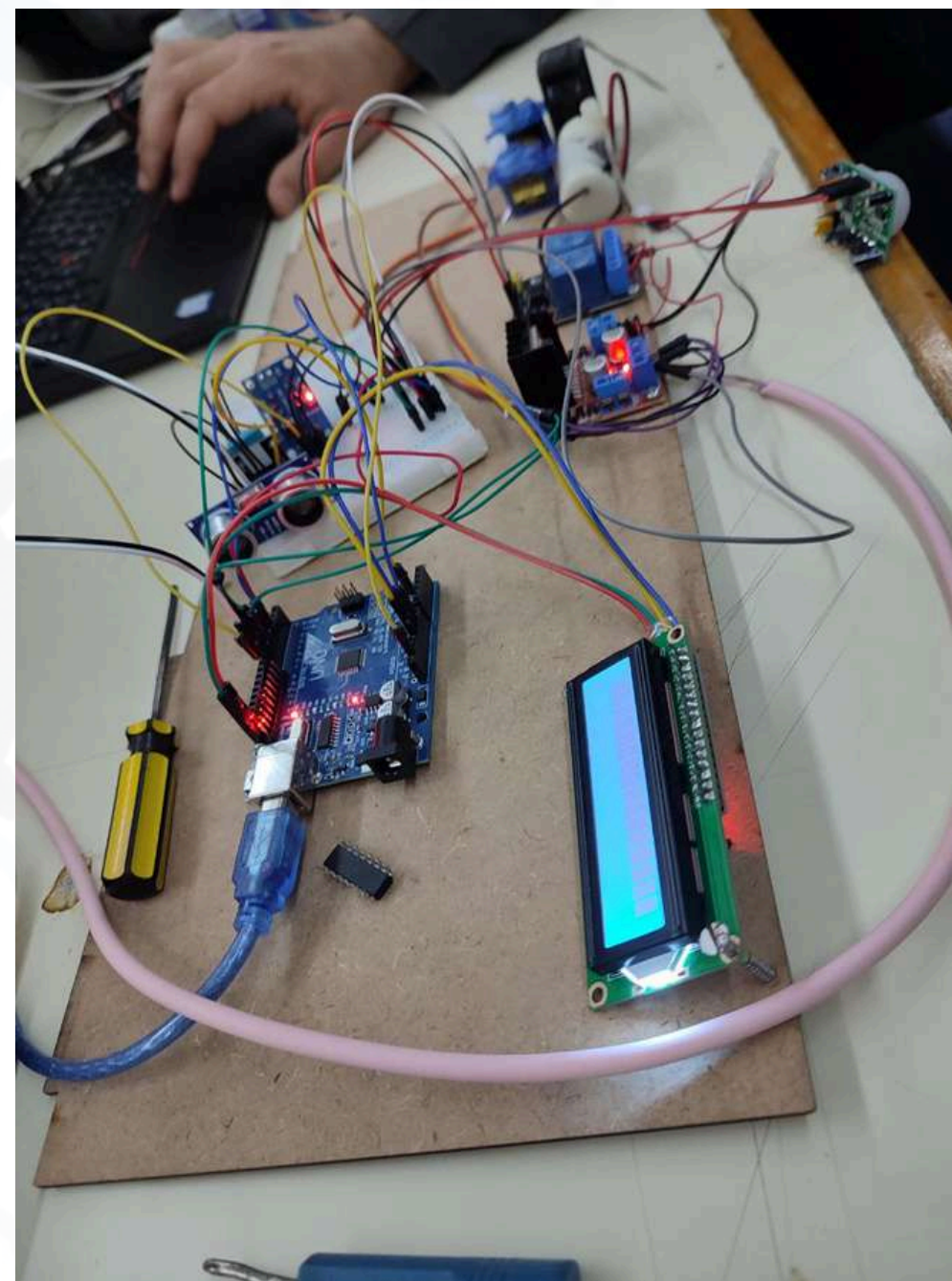
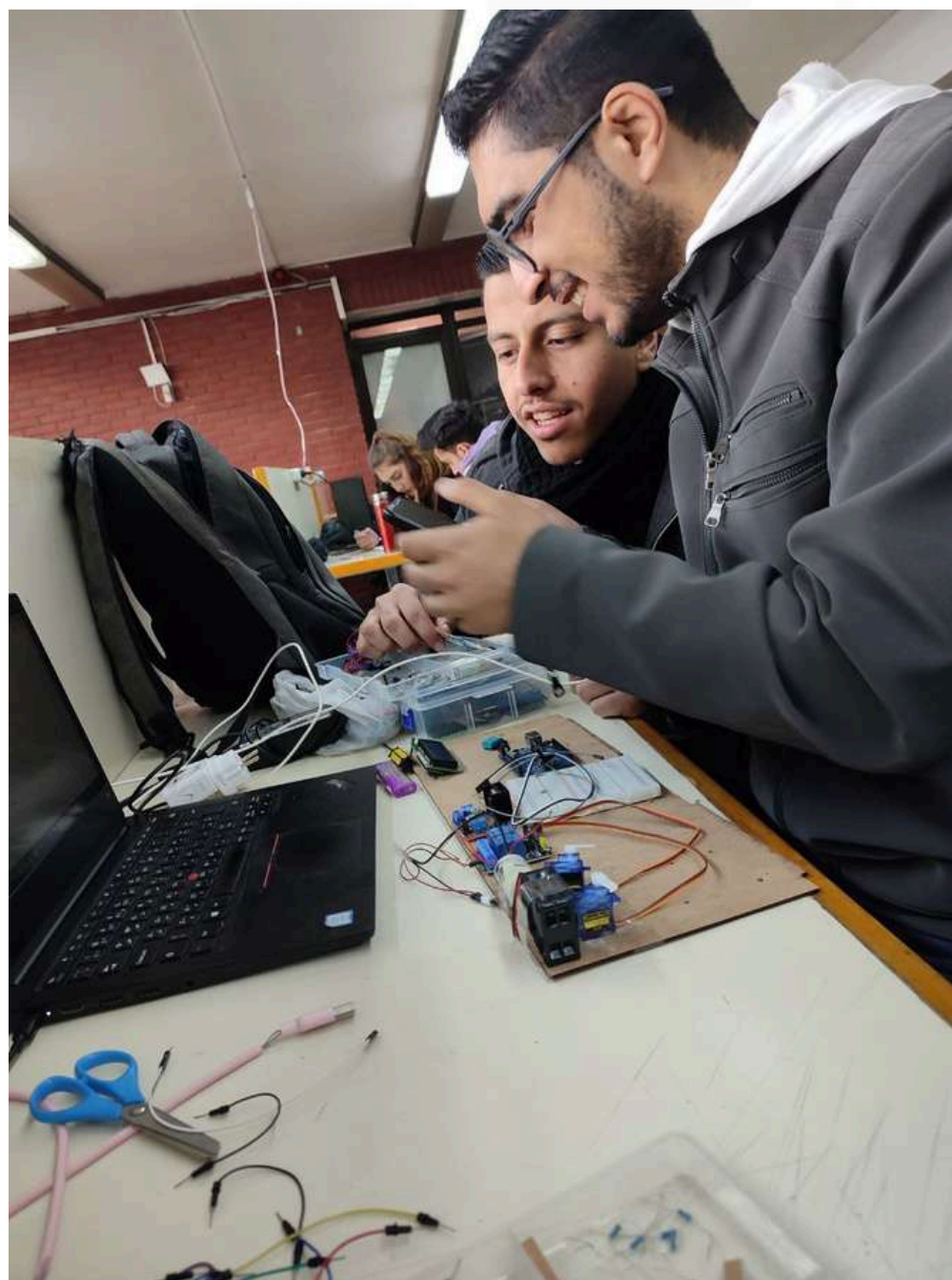
DisAdvanatges

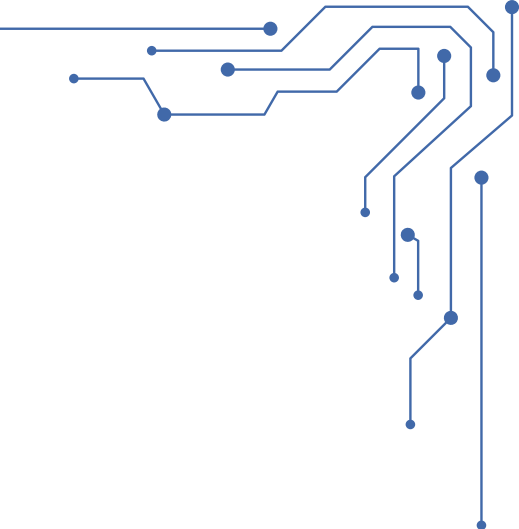
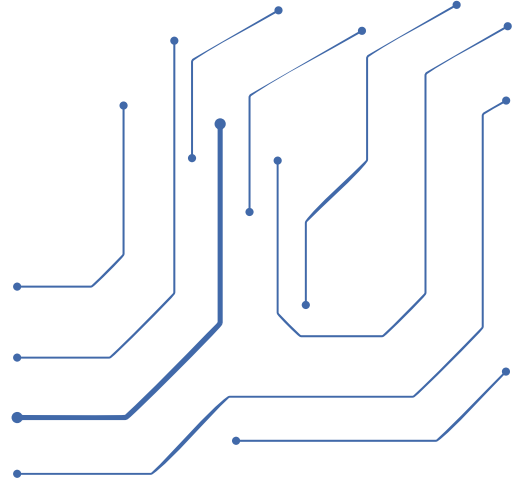
- Prototype-level system
 - Limited sensor accuracy
 - No industrial-grade components
 - Dependent on power and network availability
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Teamwork



Teamwork





Thank You

Questions?

